A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 13

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Number all possible sampling sites (eg on a grid map) **1** Select the sites using a random number generator / table – (regular shape: grid coordinates from random numbers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis **1** So the sample does not represent the whole area and conclusions are invalid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** TBT is highly toxic to marine molluscs / crustaceans and an endocrine disruptor altering growth and reproduction of oysters / whelks even at very low concentrations **1** It bioaccumulated in shellfish and could pass to humans eating them **1** It was assumed dilution would make it harmless (illustrates the need for the precautionary principle) – now prohibited and replaced by copper **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	3
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** In the low-oxygen conditions of the spoil heap iron is in its soluble reduced form and is leached into the river **1** In the river it is oxidised to insoluble iron oxide (orange sediment) – the oxidation uses dissolved oxygen so the river is deoxygenated, killing aerobic organisms **1 Control:** drainage water is passed over mesh screens where the iron oxidises and is deposited before the water reaches the river – the solid iron is removed periodically **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water	3

Do not write outside the box

Question	Marking guidance	Marks
02.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Lead-acid car batteries (recycled) **1** Lead flashing on roofs / windows **1** Radiation shielding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
02.5	1 mark for the point below. **1** They inhibit enzyme function, especially the enzymes of the nervous system / nerve cells **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 B:** Oxygen	1
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) **1** There is still a lot that is not understood about how atmospheric processes work **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** The surface warmed in the tropics heats the air above – warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes **1** Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** A fluid in a buried ground loop absorbs low-temperature heat from the ground – at the expansion valve the fluid boils at low pressure absorbing this heat **1** A compressor raises the pressure so the gas condenses, releasing heat at a higher temperature inside the building **1** The electricity only moves heat rather than creating it, so up to four times as much heat is delivered as electricity used (less if the source is colder) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Parabolic reflectors focus sunlight onto tubes of oil, increasing the energy density and reaching high temperatures **1** The heat is transferred to molten salt (potassium nitrate) stored in large insulated tanks at up to 550 °C **1** The stored heat boils water to drive steam turbines whenever electricity is required, including at night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reflectors can only focus parallel rays of direct sunlight – cloud, dust and smoke scatter light so it cannot be concentrated **1** PV cells convert diffuse as well as direct light, though output is lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **diffuse** → pollution from a wide area with no single outlet	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1 Manufacture:** extraction / processing of silicon, metals and plastics – toxic wastes (silicon tetrachloride, cadmium) **1** Land take by large solar farms competing with other uses / affecting desert flora and fauna **1** Water for cleaning panels in arid areas **1** Disposal of old panels **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 200-202; AQA spec 3.3.4.1.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** B, C and D – each loses soil faster than the 1 t ha^{-1} yr^{-1} at which it forms, so the soil is getting shallower **1** Accelerated erosion means the natural equilibrium between soil formation and erosion has been broken – field A loses only 0.8 t ha^{-1} yr^{-1}, which is less than the rate of formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2
05.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (24.0 - 9.0) / 24.0 x 100 **1** = 62.5% (accept 62 - 63%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **contour ploughing** → ploughing along the slope rather than up and down it **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 3 points below. **1 Contour ploughing:** furrows run along the contour at 90° to the slope so runoff is stopped by the furrow and loses kinetic energy – the particles it was carrying are deposited in the furrow instead of being washed away **1 Terracing:** the slope is cut into a series of narrow level fields held by retaining walls, so water slows as it crosses each field and much of it infiltrates rather than running off **1 Rows of stones:** stones laid along the contours of a gentle slope slow the runoff and trap disturbed soil behind them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Contour ploughing** → ploughing along the slope rather than up and down it • **Terracing** → cutting the slope into level steps	3
05.4	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1 Multicropping:** growing more than one crop in the field at a time, sown and harvested at different times, so some crop always covers the soil **1 Strip cropping:** wide strips of different crops, most effective at 90° to the prevailing wind, for farms that use machinery **1** Growing a long-term crop such as permanent grassland or a bush or tree crop, which does not need frequent replanting or soil disturbance **1** No-tillage cultivation / direct drilling of seed, which disturbs the soil far less than ploughing and sowing **1** Adding organic matter such as mulch or manure – surface mulch shields the soil from the wind and from rain splash, and the humus formed as it decomposes makes the particles adhere to each other **1 Controlling livestock:** lower stocking density and restricted movement so there is less trampling damage and more plant cover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4

Do not write outside the box

Question	Marking guidance	Marks
05.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Soil particles carried into rivers are deposited where the water slows, so the flow capacity of the river falls and it overflows more readily after heavy rain **1** Sediment fills reservoirs, reducing the volume of water they can store and shortening their useful life **1** High turbidity in the river reduces light penetration and smothers the river bed, damaging aquatic life **1** Wind erosion raises the level of atmospheric particulates, which worsens respiratory illness such as asthma **1 Erosion contributes to desertification:** vegetation struggles to grow on the thinner soil, which exposes more soil and reduces rainfall further **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **particulates** → fine solid particles / smoke and dust **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-166; AQA spec 3.2.5.2.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Tropospheric ozone (formed by reactions of NOx in sunlight)	1
06.2	**1 mark each** 1 mark for each of the 4 points below. **1 Primary pollutant:** released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores – NOx from high-temperature combustion **1 Secondary pollutant:** produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) **1** Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid – NOx dissolve to form nitric acid **1** The acids (sulfurous, sulfuric, nitric) are secondary pollutants – ozone itself is a secondary pollutant of NOx photochemistry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** The lowest ore purity that can be mined economically with existing technology **1** Deposit X (0.9% is above 0.4%; Y at 0.35% is part of the resource, not the reserves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.2	**1 mark each** 1 mark for each of the 6 points below. **1** X has higher grade so less rock mined, less spoil, less energy and pollution per tonne of copper **1 But X is deep:** costs rise more than proportionally with depth (sides cannot be vertical so much more rock removed; more water inflow and pumping) **1** X has hard overburden requiring blasting (cost, noise, dust) and high rainfall increasing drainage costs **1** X is far from transport infrastructure so transport costs are high **1** Y is shallow with loose overburden (cheap to remove but landslide risk so gentler slopes / larger land take), near a railway, low rainfall **1** Y's grade is below the COOG so it is not viable now but would become viable if copper prices rose or extraction technology improved (eg bioleaching) – both sulfides risk acid mine drainage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
07.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Higher market value / price of the metal **1** Improved extraction technology **1** Lower energy costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 138-139; AQA spec 3.2.3.5.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 5 points below. **1** Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier **1 Standardise the noise source:** same cars / same race or a standard test noise – same time of day, wind speed / direction, weather – same height of meter above ground **1** Take repeated readings at each distance and calculate means (and standard deviation) to assess variability **1** Use a measure such as LA10 or Leq over a set period rather than a single peak reading **1** Compare reduction in dB relative to the control at each distance – plot noise level against distance for each barrier – statistical test (t-test) to assess significance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **standard deviation** → a measure of the spread of the data about the mean • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	5
08.2	1 mark for the point below. **1** Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods **1** LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2

Do not write outside the box

Question	Marking guidance	Marks
08.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Double / triple glazing **1** Acoustic insulation in walls / roof **1** Bedrooms on the side facing away from the track **1** Dense hedges / trees and earth mounds in gardens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	
09.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** They store chemical energy that is easy to keep until it is needed, and easy to convert into the heat that is usually wanted **1** They have a high energy density, so they reach the high temperatures needed to smelt metal ores and to raise high-pressure steam for turbines **1** High energy density also makes them efficient to transport, because a small mass or volume carries a lot of energy – 75 t of aviation fuel can carry a 400 t aircraft 5600 km **1** They were often found in very abundant local deposits, so industry could grow up on the coalfield that supplied it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 75 t / 400 passengers = 0.1875 t per passenger **1** = 188 kg (accept 187 - 190 kg) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (kg) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Many deposits are too deep, too small or in places that are too difficult to reach for the technology available **1** A lot of natural gas is held in fine-grained impermeable shale, from which it cannot flow to a conventional well **1** Oil shale holds more than the world's total reserves of crude oil, but the hydrocarbons are solid and insoluble, so the rock has to be heated to melt the oil before it will flow **1** Reserves count only what can be extracted economically, so deposits that would cost more than the fuel is worth are excluded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	3
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Economic decisions are made on the cheapest option available at the time rather than on long-term supply, so the mines closed while coal was still in the ground **1** The abandoned mines have flooded and are unsafe, so the remaining coal cannot be reached even though North Sea gas will be exhausted within the next thirty or forty years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.5	**Any 5 of 7** 1 mark each for any 5 of the 7 points below, in any order, maximum 5. **1** Deep mining damages the surface through loss of habitat, the transport infrastructure it needs and subsidence above the worked seams **1** Opencast mining causes greater habitat damage because all the overburden above the coal has to be cleared **1** Spoil heaps from mining occupy land, can be unstable and are slow to revegetate **1** Oil spills from rigs pollute the sea, and oil-based drilling mud used to lubricate the drill pipes can pollute groundwater, rivers and the sea **1** Surplus gas on oil rigs is flared to reduce the risk of explosion, releasing sulfur dioxide, carbon dioxide and smoke **1** Pipeline construction damages habitat over a long, wide corridor, although careful route planning, soil replacement and replanting reduce this **1** Fracking to release shale gas uses large volumes of water, risks contaminating groundwater and can cause local earth tremors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Fracking** → pumping fluid into rock at high pressure to release gas or oil	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-190; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Stages of scientific method illustrated by ozone: identification (Antarctic UV data), hypothesis (Rowland-Molina 1974), data collection / analysis (Dobson units, satellites, balloons confirming ClO), proposed solutions, enactment (Vienna 1985, Montreal 1987) • **Evidence of success:** ozone hole area and depth stabilised and beginning to shrink – CFC concentrations in the atmosphere falling – UV levels not rising further – global ozone loss limited to about 4% • **Reasons for success:** international recognition of serious consequences (skin cancer) – agreement of nearly every country – legally binding targets with a fund for developing countries – relatively few producers / products – alternatives (HCFCs, HFCs, pump sprays, hydrocarbons) already available and cheap – no lifestyle change needed • **Limits:** persistence of CFCs means recovery takes decades – some ODSs still used – HFC replacements are greenhouse gases • **Climate change contrast:** greenhouse gases come from almost every human activity (energy, transport, agriculture, industry) in every country – fossil fuels underpin economies so control requires major lifestyle and economic change • Agreements (Kyoto, Paris) have weaker / non-binding targets, disputes between developed and developing nations – emissions still rising – natural fluctuations, time delays and tipping points make cause-effect less clear to the public	9

Do not write outside the box

Question	Marking guidance	Marks
	• **Judgement:** ozone control was a technical substitution problem – climate change is a systemic one, so the ozone success is only partly transferable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • The converse of any point, ie the same comparison written the other way round. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **persistence** → does not break down / stays in the environment for a long time • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	
	Model answer - Level 3 (9 marks)	
	Ozone depletion followed the scientific method almost textbook-fashion: Antarctic UV and ozone measurements identified a problem, Rowland and Molina's 1974 hypothesis explained it, Dobson-unit, satellite and balloon data confirmed chlorine monoxide in the stratosphere, and the Montreal Protocol enacted a solution. It has largely worked: CFC concentrations are falling, the ozone hole has stabilised and is starting to shrink, and global ozone loss was limited to about 4%.	
	Success came from particular conditions. The consequence - skin cancer - was clear and frightening, nearly every country agreed, targets were legally binding with a fund to help developing countries, there were relatively few producers and products, and alternatives (HCFCs, HFCs, hydrocarbons, pump sprays) already existed, so substitution cost little. Even so recovery will take decades because CFCs are so persistent, some ozone-depleting substances are still in use, and the HFC replacements are themselves greenhouse gases.	
	Climate change has none of these advantages. Greenhouse gases come from almost every activity - energy, transport, agriculture, industry - in every country, and fossil fuels underpin entire economies, so control means changing lifestyles and economies rather than swapping one chemical for another. Kyoto and Paris set weaker, largely non-binding targets, developed and developing nations dispute responsibility, and natural fluctuations, time lags and tipping points make cause and effect less obvious to the public. Emissions are still rising. The ozone story shows international agreement can work, but it was a technical substitution problem; climate change is systemic, so its lessons transfer only partly.	
	Weaker answer - Level 2 (4 marks)	
	Ozone depletion was caused by CFCs. Scientists found the hole over Antarctica and the Montreal Protocol banned CFCs, so the ozone layer is now recovering. This was successful because everyone agreed and there were alternatives to CFCs. Climate change has not been tackled as well. There have been agreements like Kyoto and Paris but carbon dioxide is still going up. This is because fossil fuels are used everywhere and countries do not agree. So ozone depletion was solved better than climate change.	

Do not write outside the box

Question	Marking guidance	Marks
	What makes the difference	
	• Level 2 gives the right outline but Level 3 supports it with evidence (falling CFC levels, 4% loss, Dobson units, ClO detection) and explains each reason for the difference.	
	• Level 3 acknowledges the limits of the ozone success (persistence, HFCs) - balance is expected at the top level.	
	• The final judgement in Level 3 characterises the difference (substitution vs systemic problem); Level 2 just restates the question.	
10.2	1 mark for the point below. **1** Vienna Convention for the Protection of the Ozone Layer - no legally binding goals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Ozone Layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113, 105-106; AQA spec 3.2.1.3, 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility – DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • **Reducing impact:** restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • **Alternatives:** cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • **Trade-offs:** yield, cost, human health (organophosphates), resistance evolution, pollinator services – risk:benefit – sustainability and the circular economy	25
	• **Judgement:** no pesticide is perfectly specific – combination of property-based selection and integrated control is most sustainable *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • **How properties predict behaviour:** dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • **Control based on properties:** persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones – systemic pesticides and seed coatings • **Heavy metals:** lead uses discontinued, storage at high pH to reduce solubility – mercury chemical form – radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays) – oil properties (floats, insoluble) exploited by booms and skimmers	25

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	– density in cyclone separators • **Principles:** CPA and CGM based on properties – emission location / timing – ALARA and BATNEEC – polluter pays – precautionary principle • **New technologies:** satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • **Evaluation:** prevention better than cure – limits of remediation – cost-effectiveness – contrast success of ozone (Montreal) with climate change – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	
	Model answer - Level 5 (22 marks)	
	Pollutants cause damage in proportion to how much reaches sensitive organisms and how harmful it is when it gets there. Both depend on properties, so knowing a pollutant's properties lets us predict where it will go and which control will work.	
	State of matter and density decide dispersal. Solid particulates settle near the source unless they are very small: PM10 and smaller stay suspended, travel far and penetrate lungs, so smoke is controlled at the point of release with electrostatic precipitators, cyclones and bag filters, and smokeless zones prevent its production. Gases disperse widely, and dense gases such as the hydrogen cyanide released at Bhopal hug the ground, so location and emission height matter. Persistence decides whether dilution is acceptable. Sewage and other biodegradable wastes break down, so treatment plus dilution in a river or the sea is sufficient; organochlorine pesticides, CFCs, heavy metals and long-lived radioisotopes are not broken down, so they travel, accumulate and must be contained, banned or replaced - DDT and CFCs were both controlled by substitution once their persistence was understood.	
	Solubility governs the pathway. Water-soluble nitrate leaches into rivers and aquifers, so it is controlled by nitrate vulnerable zones, fertiliser timing and buffer strips; liposoluble pollutants such as DDT and methyl mercury pass into fat and are biomagnified along food chains, as at Minamata, so even low concentrations in water are dangerous and the only effective control is to stop release. Adsorption on clay immobilises caesium and metals, which is why radioactive discharges are monitored in sediments and why activated carbon and polymer adsorbents are used deliberately in effluent treatment.	

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	Toxicity, specificity and mode of action set acceptable doses and identify species at risk. Neonicotinoids are toxic to insects but not vertebrates because they cannot cross the vertebrate blood-brain barrier, which explains both their popularity and their effect on bees; pyrethroids replaced organochlorines because they are less persistent. For mutagens and carcinogens such as ionising radiation there is no safe dose, so exposure is kept As Low As Reasonably Achievable with shielding, distance, time limits and monitoring. Reactivity explains secondary pollutants - sulfur dioxide oxidised to sulfuric acid, nitrogen oxides and hydrocarbons forming ozone and PANs in photochemical smog - so control targets the primary emissions with flue-gas desulfurisation and catalytic converters. Synergism between smoke and sulfur dioxide, or ozone and sulfur dioxide, means limits must consider pollutants together.	
	Properties interact with the environment: temperature inversions trap pollutants, currents disperse them, pH and oxygen alter solubility. Critical Pathway Analysis combines the two to predict where a discharge will concentrate and to focus monitoring, and Critical Group Monitoring checks the most exposed people. These principles have reduced smoke, sulfur dioxide, lead and organochlorines dramatically in developed countries. Their limits are pollutants with diffuse sources or that underpin economies - nitrate, carbon dioxide - where knowing the properties is not the problem; changing behaviour is. Even there, the precautionary principle applies the same logic: predict the properties of a new substance from similar ones and control it before release, because prevention is always cheaper than remediation.	
	What makes the difference	
	• Every paragraph is organised around a **property**, each linked to a prediction and a named control - the essay answers the question throughout rather than describing pollutants.	
	• Examples span the specification (smoke, acid rain, pesticides, heavy metals, radiation, nutrients, oil), with quantitative and named detail (Bhopal, Minamata, PM10, ALARA), which is what Level 5 breadth means.	
	• The final paragraph evaluates: where the approach has worked, where it has not and why, and reaches a justified conclusion.	
	• A Level 3 version of this essay would describe several pollutants and their effects accurately but only occasionally connect a property to a control, and would end without a judgement.	

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